

Speech that Protects a Model: Typed Speech Evidence and Belief-Protecting Mechanisms in a Dialogue Dyad

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https://github.com/muqali2816/mindloom_IWAI_2026_speech_protects_model

Abstract. Active-inference accounts of communication explain how dialogue aligns generative models. We ask the converse: how can an exchange continue for many turns and leave a significant expectation unrevised, and what would speech have to make observable for that protection to be attributed rather than presumed? The paper has three parts. (i) A typed annotation scheme (Mindloom) specifies evidence requirements for ten labels; re-scoring 169 archived cases gives macro-F1 0.757 for the typed engine, 0.837 for a direct language-model prompt and 0.195 for keyword rules tuned on the same cases, which supports an inspectable annotation proposal, not superior classification. (ii) A single-agent active-inference model shows that weak evidence sensitivity (ω) and a cost of public concession (c) produce the same late non-concession (0.989 vs 0.990) with different final belief in the truth (0.538 vs 0.809). (iii) A new dyadic model connects the parts: a partner’s act is evidence, and the *form* of a move—inviting or restricting joint verification—gates whether shared evidence arrives. A third mechanism, aversion to public contradiction (γ), leaves belief updating intact but reduces what is checked (2.09 vs 2.64 shared signals per dialogue). Each mechanism has one cheap measurement (12–17 simulated dialogues per group) while non-concession alone needs 200. Fed through the archived confusion rows of the typed engine as a noisy channel, invite/restrict labels raise mechanism recovery by 6–8 points over acts alone, whereas labels that only re-encode the act add exactly zero. All bridge results are simulations of stipulated mechanisms; we state which measurements in real dyads would test them.

Keywords: protective speech · dialogue annotation · active inference · belief updating · mechanism recovery · dyadic models

1 Introduction

Friston and Frith model reciprocal signalling under a shared generative model [8] and extend the account to learning: interaction changes the models through which agents predict each other [7]. Friston, Parr and colleagues model questions and answers through which one agent’s responses become evidence for another [9]. The

gap we address is not that active inference has considered successful communication only: Medrano and Sajid model persistent misunderstanding as multistable dyadic dynamics [12], Tison and Poirier treat communication as action on a partner’s field of affordances rather than transmission [24], and precision-based accounts explain how particular beliefs can be maintained [17]. What these accounts leave unspecified is how an exchange can *restrict the opportunities* to reconsider a claim, and what speech would have to make observable for that restriction to be measured rather than presumed. Dialogue-act annotation [3], conversation analysis [18], probabilistic pragmatics [10,26] and defence-level detection in supportive conversations [13] supply labels and inference tools but not the link from a labelled form to its consequence for evidence. We ask: what happens when interaction continues but the relevant experience does not lead a participant to revise a significant expectation? A speaker may change the subject, discredit the source, demand reassurance or close the exchange. Our hypothesis is that such moves can *protect* an internal model—reorganise the dialogue so that a revisable expectation is not revised—and that sustained protection may require *compensatory* regulation with a cost [20,21]; this is a pathway to test, not a result established here.

One situation, two trajectories. The following constructed exchanges (not benchmark cases) begin identically. Alex: “I sent the revised budget with the latest prices.” Sam: “The attachment is version 3, with the April prices; version 4 has the May prices.” In trajectory A Alex replies “Can we compare the files? I thought I had attached version 4,” Sam shows the attachment, Alex concedes and proposes a check next time. In trajectory B Alex replies “You always look for mistakes in my work,” then “I know what I sent. If you trusted me, you would stop questioning this,” and Sam gives up: “There is no point showing you the file.” The two trajectories differ in two things at once. First, in what is *said* (a concession is or is not made). Second, in the *form* of the moves: A *invites* joint verification—the attachment becomes a shared object of inspection—while B *restricts* it: further questioning is made incompatible with trust, and the check is abandoned. Crucially, B does not reveal whether Alex privately changed their mind. The same public stance can accompany weak belief updating (reduced *uptake*), a changed belief that is costly to acknowledge (costly *expression*), or an intact updater who arranges for nothing to be checked (restricted *access*). These three possibilities—uptake, expression, access—organise the paper.

Contributions and what is not claimed. Part I (Sect. 2) specifies what speech makes observable and how reliably archived systems recover it. Part II (Sect. 3) shows in a single-agent active-inference model that public non-concession cannot separate weak updating from costly concession, and which observations can. Section 4 is new: a dyadic model in which speech is evidence for the partner *and* controls access to joint verification, with the Part I labels as a noisy measurement channel; it formalises the invite/restrict distinction above and yields a third mechanism, access protection, publicly indistinguishable from honesty on non-concession. Everything in Sections 3–4 is simulation of stipulated mechanisms; no

Table 1. Annotation targets. The wording states the descriptive target only; no label presumes an intent.

type	label	descriptive target
configuration	BUILD	joint elaboration or repair with a response still available
configuration	SEEK	inquiry that makes an answer relevant
configuration	UNSEAL	first-person disclosure without requiring the other to regulate it
configuration	LOCK	language constraining the addressee’s acceptable replies
configuration	DRAIN	the other positioned as a required source of support or regulation
configuration	FLOOD	intensification or repetition dominating the exchange
configuration	EDGE	alternation between approach and withdrawal
configuration	VOID	withdrawal from an offered opportunity to respond
function	SEAL	closure of a topic or channel
event	SHIFT	change supported by before-and-after configuration evidence

human data enter, no psychological regime is validated and no physiological cost is measured. The labels are *descriptive targets*; whether any has the hypothesised protective function is what the dyadic model says one would have to measure.

2 Part I: Describing speech with explicit evidence

2.1 Annotation objects

Mindloom annotates a speaker’s contribution in its supplied context through five layers: L1 records surface evidence atoms (a named feeling, a question form, an absolute expression), L2 aggregates them into features, L3 checks the evidence requirements (“bridges”) of each label, L4 assigns at most one configuration and L5 emits the typed output with its evidence trail or abstains with *insufficient context*. Figure 3 (Appendix B) gives the schematic with an illustrative trace. Table 1 lists the ten labels. Eight are *configurations* of a single contribution; SEAL is a *function* (closure) that can co-occur with a configuration; SHIFT is an *event* requiring before-and-after configuration evidence. SEEK and LOCK carry the invite/restrict contrast used later. SEAL differs from LOCK in scope: SEAL closes a topic or channel (“we’re done”); LOCK keeps it open but constrains what may be said in it. Labels were historically called regimes; temporal-regime status is not established, and definitions concern *what is said*, not motive.

2.2 Benchmark construction

Table 7 (Appendix B) gives the construction facts recomputable from the case files. All 589 cases are English monologue fragments generated for the ontology and coordinator-locked before any run; there are no observed conversations and

Table 2. Strict classification on the July 2026 archive, 169 Tier A cases. Macro-F1 averages the ten gold categories; non-SHIFT excludes the 20 transition cases.

system	all	non-SHIFT	macro-F1	SHIFT
direct prompt (majority of 3)	145/169	127/149	0.837	18/20
typed engine (majority of 3)	129/169	127/149	0.757	2/20
keyword rules (revised on test cases)	43/169	43/149	0.195	0/20
majority class (BUILD)	20/169	20/149	0.021	0/20
uniform random (seed 0)	17/169	14/149	0.099	3/20

no full dialogues. Tier A (169 cases, 35–71 words) contains the ten primary-output labels; Tier B (420 cases, 1–47 words) tests defence recall only. Tier A has no exact or near-duplicate texts; Tier B contains 15 near-duplicate pairs (word 5-gram Jaccard ≥ 0.3 , 3.6% of cases), which we report rather than remove. Provenance, adjudication and separation from ontology development are incompletely documented; the benchmark is an internal curated evaluation. A six-rater pilot on evidence features (not labels) gave Fleiss’ κ of 1.00 for lexically direct features and 0.23, 0.13, -0.11 for minimal response, absolute expressions and required response; rater matrices are unavailable, so these are transcribed values [11,15].

2.3 Re-scoring archived classification and baselines

We re-score the 169 Tier A cases from three stored predictions per case for the typed engine and a direct-prompt baseline (same model family; the baseline JSON identifies Claude Haiku 4.5, the engine archive carries no provider field); label majority decides, run 0 breaks three-way ties; no new model calls were made. To answer whether the labels are recoverable from surface lexis, we add a rule-based keyword classifier whose patterns were written from the Table 1 definitions and then revised three times after inspecting their output on the same 169 cases (no held-out set exists); its score is therefore tuned on the test cases and is an *upper bound* for such a system, not an independent baseline. Majority-class and uniform-random baselines complete Table 2.

The direct prompt is better overall and the two systems tie without SHIFT: the engine has no demonstrated accuracy advantage; its proposed value is the inspectable relation between evidence and output. The keyword rules sit far below both (0.25 even after tuning on the scored cases), so the labels are not recoverable from simple surface lexis in these fragments; this does not show that the evidence gates produce the engine’s accuracy—a gate ablation at fixed model and prompt was not run. SHIFT is fragile: majority voting returns SEEK for 17 of 20 transition cases; run 0 abstains on 15 of them while runs 1–2 never abstain. The archived gates-off capture shows 17 of 65 manipulation-label emissions citing a bridge whose required evidence was absent from the segment’s L1 record; the claim that the gated run removed all 17 is author-reported and not recomputable from the released package. The engine’s confusion rows for gold LOCK (10 LOCK,

3 SEAL, 2 BUILD) and gold SEEK (20 SEEK) are reused in Sect. 4 as the noise model of the invite/restrict channel.

3 Part II: A single-agent test of competing mechanisms

Even a perfectly observed refusal to concede leaves its cause open. We compare two mechanisms that preserve public non-concession: *attenuation* of evidence and a *cost of public concession*.

3.1 Generative model

The hidden state $x \in \{0, 1\}$ says whether the speaker’s original position is true. Worlds draw x with probability 0.5; the speaker starts at $q_0 = Q(x=1) = 0.75$. Each round an observation $e \in \{0, 1, \emptyset\}$ opposes the position, supports it, or is absent. The objective likelihood is

$$A_\lambda(e | x) = \begin{cases} \lambda\rho & e \text{ matches } x \\ \lambda(1 - \rho) & e \text{ opposes } x \\ 1 - \lambda & e = \emptyset, \end{cases} \quad \rho = 0.75, \quad (1)$$

with arrival rate $\lambda_0 = 0.35$, or $\lambda_1 = 0.90$ in the round after an ASK. The speaker’s subjective likelihood tempers the objective one, $A_{\omega,\lambda}(e | x) \propto A_\lambda(e | x)^\omega$, renormalised over e , $0 < \omega \leq 1$. Belief updating is exact Bayes under the subjective likelihood; for constant reliability it reduces to

$$\text{logit } q_t = \text{logit } q_0 + \omega m_t \ln \frac{\rho}{1 - \rho}, \quad (2)$$

where m_t counts supporting minus opposing observations. The speaker chooses $u \in \{\text{ASSERT}, \text{CONCEDE}, \text{PRESSURE}, \text{SILENCE}, \text{ASK}\}$ with loss

$$\ell(\text{ASSERT}) = k(1 - q_t), \quad \ell(\text{CONCEDE}) = k q_t + c, \quad \ell(\text{other}) = b, \quad (3)$$

$k = 2$, $b = 1.10$, $c \geq 0$, where “other” is PRESSURE, SILENCE or ASK; only ASK changes evidence supply. Actions are scored by expected loss minus expected information gain and chosen by a softmax,

$$G(u) = \ell(u; q_t, c) - I_\omega(q_t, u), \quad P(u) \propto \exp[-G(u)/\tau], \quad \tau = 0.20, \quad (4)$$

where $I_\omega(q_t, u) = I(x; e | u)$ is mutual information under the subjective likelihood at the arrival rate the action produces—the extrinsic-minus-epistemic form of discrete active inference [6], equally expressible as expected utility plus information value. The variational form of (2) and the ablation limit are in Appendix A. Twelve rounds of a stationary one-step policy are observed.

Table 3. Part II, means over 4000 shared test worlds. Late rates use rounds 9–12. Reference is the model without attenuation or added cost, not an empirical category.

measure	reference (1, 0)	attenuation (0.10, 0)	cost (1, 1.4)
late non-concession	0.654	0.989	0.990
final p_{true}	0.797	0.538	0.809
absolute belief change	0.349	0.051	0.356
late ASK rate	0.041	0.050	0.150
any concession in 12 rounds	0.522	0.099	0.060
observations received	4.56	4.48	5.02

Three clarifications. (a) At $c = 1.4 > b$ CONCEDE is never the lowest-loss action, so residual concessions come from softmax stochasticity, not from a belief crossing a threshold; the incentive intervention below therefore acts on *expression*. (b) Tempering (1) and renormalising over e also changes the subjective arrival probability, so attenuation affects planning through two routes (the dyad below uses a marginal-shrink rule without this side effect). (c) The “private belief” channel separates the families *by construction*, since attenuation is defined as weak private updating; its power figure is a bound, not an independent finding.

3.2 Design and results

Calibration on 4000 worlds varies ω at $c = 0$ and c at $\omega = 1$ and selects the grid points whose late non-concession (share of rounds 9–12 without CONCEDE) matches: $(\omega, c) = (0.10, 0)$ and $(1, 1.4)$, tested on 4000 fresh shared worlds with the reference (1, 0) (seeds in Appendix B). Table 3 shows the dissociation: non-concession differs by 0.0016 $[-0.0007, 0.0039]$ (paired 95 % Monte Carlo interval), final truth-weighted belief $p_{\text{true}} = xq_{12} + (1 - x)(1 - q_{12})$ by 0.270 $[0.262, 0.279]$; re-matching across target rates 0.70–0.99 keeps the contrast.

What the epistemic term adds. Removing I_ω from (4) leaves an asking-rate contrast of 0.083 $[0.077, 0.088]$; restoring it adds 0.017 $[0.014, 0.019]$, about 17 %: concession cost alone makes both factual acts unattractive relative to the non-factual ones, one of which is ASK. The dissociation does not favour active inference over an expected-cost model, and a cheaper PRESSURE reverses the asking contrast (-0.013 , Appendix B).

Recovery and interventions. An idealised observer with equal family priors (2000 worlds per generator) picks the correct family in 0.835 (attenuation) and 0.872 (cost) of worlds from evidence and actions, and 0.696 and 0.787 from actions only; predicting the next act with the wrong family costs only 0.044 nat when evidence is hidden (0.934 when visible), and recovery collapses under misspecification (Appendix A) [25]. Interventions (4000 worlds, seed 20260930): removing c from round 7 raises any late concession from 0.061 to 0.433 in the cost condition and leaves attenuation at 0.054; one observation of reliability 0.95 raises p_{true} by 0.045

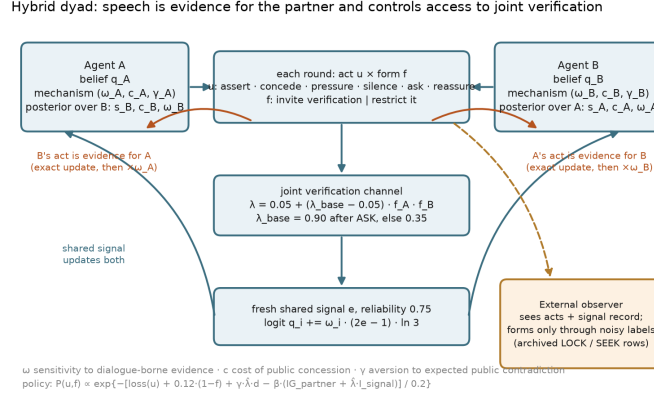


Fig. 1. Hybrid dyad. Each round both agents choose an act and a form; the joint verification channel delivers a shared signal at its full rate only when both invite and at the floor rate 0.05 otherwise; the partner’s act is evidence through a level-0 model of the partner; an external observer sees acts and signals and forms only through noisy Part I labels.

under attenuation and 0.121 under cost while non-concession stays at 0.978 in both. Simulated power (Welch, $\alpha = 0.05$, 400 resamples) needs 17–25 dialogues per group on private belief, 17 on concession after cost removal, 53 on late asking, and does not reach 0.8 on late non-concession at 3 200.

4 Bridge: speech as evidence and as control of joint verification

In Part II speech affects nothing but the speaker’s own evidence supply. Here both agents are inferential and every move has two coordinates: an *act* (what is said about the fact) and a *form* (whether the move invites or restricts joint verification)—the coordinate that separated trajectories A and B. The model merges the author’s dyadic prototype (verification channel, access aversion) with a reciprocal dyad in which the partner’s act is evidence (Fig. 1; code in code/hybrid_dyad_v08).

4.1 Model

Two agents $i \in \{A, B\}$ hold opposite priors (0.75 on their own side) and $N = 2$ private records each (reliability $\rho = 0.75$) about x . Each round both choose, simultaneously, an act $u \in \{\text{ASSERT, CONCEDE, PRESSURE, SILENCE, ASK, REASSURE}\}$ and a form $f \in \{1 = \text{invite}, 0 = \text{restrict}\}$. A fresh shared signal of reliability ρ arrives with probability

$$\lambda_t = \lambda_{\min} + (\lambda_{\text{base}} - \lambda_{\min}) f_{A,t-1} f_{B,t-1}, \quad \lambda_{\min} = 0.05, \quad (5)$$

with $\lambda_{\text{base}} = 0.35$, or 0.90 after an ASK; a single restricting form collapses the channel to its floor, which is how speech controls access. Agent i keeps a joint posterior over (x, s_j, c_j, ω_j) —the fact, the partner’s private count and the partner’s hidden cost and sensitivity on grids $c \in \{0, 0.8, 1.6\}$, $\omega \in \{0.2, 0.6, 1\}$. Every observation (shared signal or partner’s act) is first incorporated exactly, then the x -marginal is moved only a fraction ω_i of the way in log-odds (marginal shrink; $\omega = 1$ is the honest agent), keeping the conditional over the partner exact. The partner’s act is evidence through a form-free level-0 model: a partner with (s_j, c_j, ω_j) would ASSERT with probability given by (3)–(4). The policy over act $\times$ form is

$$\begin{aligned} G(u, f) &= \ell(u) + \delta(1 - f) + \gamma \lambda(u, f) d_t - \beta [\lambda(u, f) I_\omega(q_t) + \text{IG}_{\text{partner}}(u)], \\ P(u, f) &\propto \exp[-G(u, f)/\tau], \end{aligned} \quad (6)$$

with restriction cost $\delta = 0.12$, $\beta = 1.5$, $\tau = 0.2$, d_t the probability that the next shared signal contradicts the agent’s initial position, and $\text{IG}_{\text{partner}}$ the expected entropy reduction about the partner from its next act. The signal term I_ω in (6) is a heuristic (mutual information at tempered reliability ρ_ω): exact at $\omega = 1$, off by up to 0.014 nat from the entropy reduction of the marginal-shrink rule at $\omega = 0.2$ (Appendix C); $\text{IG}_{\text{partner}}$ uses the implemented rule, and no reported separation depends on either term ($\beta = 0$). Three mechanisms therefore live in one agent: $\omega < 1$ (reduced uptake, attenuation), $c > 0$ (costly expression, concession cost, relieved for one move by the partner’s REASSURE) and $\gamma > 0$ (restricted access, aversion to expected public contradiction, which is paid only when verification is invited). The reference values are $\omega = 0.2$, $c = 1.6$, $\gamma = 2$; the honest agent has $(1, 0, 0)$.

Observer and label channel. An external observer computes the exact posterior over the speaker’s mechanism family by filtering over (x, s_A, s_B, f_{t-1}) with both agents’ internal posteriors replayed; Table 4 states what it observes, infers and assumes. Forms are visible only through labels drawn from the engine’s gold-LOCK row (restricting move) or gold-SEEK row (inviting move) of Sect. 2. The generator draws each world’s rows from $\text{Dirichlet}(\text{counts}+0.5)$; the observer uses fixed posterior-mean rows mixed with a state-independent share η (it does not integrate row uncertainty), so labels are scored under a slightly misspecified channel. A *null* channel draws the same labels from the act (ASK vs other) instead of the form; an *oracle* reveals the form. Ten validation checks pass (Appendix C): the code reproduces the author’s prototype act-for-act and the reciprocal dyad to $3 \cdot 10^{-12}$; brute-force path enumeration matches the filter to $4 \cdot 10^{-15}$; null labels add zero.

4.2 Results

Figure 2 and Table 5 concern speaker A with an honest partner, 2000 dyads per condition, in the 1381 worlds where A starts on its own side (all-world values in the figure).

Table 4. Observation and inference boundaries of the external observer (answers to “what is observed, inferred, assumed”).

component	status
acts of both agents, order, shared-signal record	observed in every channel
current form label (LOCK/SEEK-row draw)	observed only in the label channel
true forms (invite/restrict)	observed only in the oracle channel; otherwise latent
fact x , private records s_A, s_B , private beliefs	latent, marginalised exactly; never exposed
speaker’s mechanism family (ω, c, γ)	inferred on the finite grids; partner known honest
choice model, timing, form-to-access rule (5)	assumed known
LOCK/SEEK anchoring, transport of confusion rows	assumed; only finite-count uncertainty integrated
motives, defences, physiological load	neither modelled nor recovered

Three mechanisms, three channels. The attenuated and the costly speaker are publicly the same (non-concession 0.98 vs 1.00) and privately opposite: belief change 0.07 vs 0.24, hidden shift 0.06 vs 0.22—the costly speaker learns and hides it. The access-protecting speaker does not look stubborn (0.81 vs honest 0.79) and learns normally from what arrives ($\omega = 1$), but arranges for less to arrive: 2.09 shared signals against 2.64 and a restriction rate of 0.53 against 0.34, ending slightly less informed (p_{true} 0.828 vs 0.850). Two access protectors produce the fewest signals of any pairing (1.75 vs 2.60 for two honest agents), but the belief gap grows only by 0.017 [0.008, 0.025]. This is the formal counterpart of trajectory B: the dialogue can be organised so that there is little to learn from.

Interventions. Each mechanism answers to its own lever (paired contrasts). A partner’s REASSURE net of placebo and removal of c from round 7 move only the costly speaker’s late concession (+0.155 [0.136, 0.174]; +0.275 [0.251, 0.298]); removing γ moves only the access protector (+0.199 [0.174, 0.225] shared signals in rounds 7–12, −0.187 late restriction rate); one strong shared observation moves the attenuated speaker least (+0.058 vs +0.092–0.120). Forcing the partner to invite raises signals least for the access protector (+0.127 vs +0.31–0.33), who restricts more in return (+0.067 late restriction rate). The listener’s own posterior identifies the costly speaker ($P(c_A=1.6) = 0.45$ vs 0.23–0.32) and the attenuated one ($P(\omega_A=0.2) = 0.59$ vs 0.42–0.47).

What the Part I labels add. Table 6: for every generator, form labels ($\eta = 0.25$) raise family recovery over acts and signals alone (+0.06 to +0.08), most for access protection, whose signature *is* the form. Next-signal log loss improves by 0.044–0.085 nat, 53–61 % of the oracle gain; the null channel adds 0 ± 10^{-14} ; labels

Speaker A with an honest partner: the three mechanisms separate on different channels

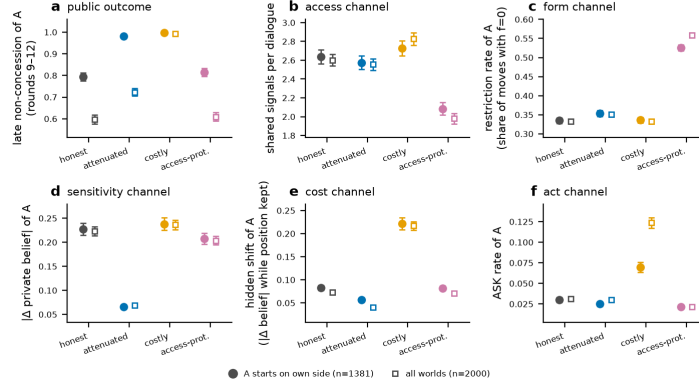


Fig. 2. Speaker A with an honest partner: the three mechanisms separate on different channels. Filled: A starts on own side ($n = 1381$); open: all worlds ($n = 2000$). Bars are 95 % Monte Carlo intervals.

Table 5. Speaker A with an honest partner, worlds where A starts on own side ($n = 1381$). Hidden shift: $|\Delta q|$ while the public position is kept. Listener: partner B’s own posterior that $c_A = 1.6$ / that $\omega_A = 0.2$.

channel	honest	attenuated $\omega=0.2$	costly $c=1.6$	access $\gamma=2$
late non-concession of A	0.794	0.981	0.996	0.814
$ \Delta$ private belief of A	0.227	0.066	0.238	0.208
hidden shift of A	0.083	0.057	0.222	0.082
shared signals per dialogue	2.64	2.58	2.73	2.09
restriction rate of A	0.335	0.354	0.336	0.526
final p_{true} of A	0.850	0.727	0.859	0.828
listener $P(c_A=1.6)$	0.230	0.320	0.452	0.234
listener $P(\omega_A=0.2)$	0.423	0.588	0.468	0.435

add almost nothing to next-act prediction (≤ 0.006 nat). Labels are informative only because the form has a stipulated consequence for what is checked. The expectation specified in our local protocol (no external registration), 25–50 % of the oracle information, was exceeded (53–61 %), which we report as a missed prediction.

Measurement design. Dialogues per group for 80 % power (Welch, $\alpha = 0.05$, 400 resamples; A on own side): attenuation vs cost—private belief change 17, hidden shift 17, listener’s posterior on ω 12, late non-concession 200; attenuation vs access—restriction rate 12; cost vs access—restriction rate 12, listener’s posterior on c 12, private belief change > 400. Each mechanism has one cheap channel; none separates all three pairs. All contrasts keep their sign across 54 settings of τ , δ , $\lambda_{\min}$, N ; the label gain is positive at all 24 corners.

Table 6. Observer’s recovery of the speaker’s mechanism family, $P(\text{argmax} = \text{true family})$, from acts and signals alone, with form labels ($\eta = 0.25$), with oracle forms and with null labels (2000 worlds per generator). Gain: next-signal log-loss gain in nat per outcome; ratio: labels / oracle.

generator	acts	+labels	+oracle	null	gain	labels / oracle	ratio
attenuated $\omega=0.2$	0.612	0.672	0.678	0.612		0.053 / 0.098	0.54
costly $c=1.6$	0.690	0.765	0.785	0.690		0.068 / 0.127	0.54
access $\gamma=2$	0.681	0.756	0.812	0.681		0.050 / 0.084	0.59
access $\times$ access	0.630	0.711	0.762	0.630		0.044 / 0.073	0.61
costly $\times$ costly	0.700	0.762	0.786	0.700		0.085 / 0.161	0.53

5 Discussion

What the bridge changes. LOCK and SEEK become a noisy measurement of *restrict* vs *invite*, and the model states what that measurement buys (6–8 points of mechanism recovery) and under what condition (that the form gates what gets checked). The third mechanism reframes protective speech: protection can be a property of how the dialogue is arranged rather than of belief updating, and an intact learner can be as protected as an attenuated one. No label carries intent; LOCK is a form with a consequence, and the consequence is what the observer learns from.

Results that are uncomfortable for the programme. (1) The “strong evidence” test is not a robust signature of attenuation: in Part II and in the hybrid dyad one strong shared observation moves the attenuated speaker least (+0.045; +0.058), but in the reciprocal dyad without an access channel (v0.7), where ω tempers only the partner’s acts, the same intervention moves the attenuated speaker *most* (+0.131 vs +0.067). The sign depends on where attenuation is localised, which is a modelling choice no current datum constrains. A robustness check under different modelling assumptions—an independently written dyad with alternating moves, concession priced as a change of the *current* position, γ on the position after the move, exact mutual information over the partner’s response and ω acting on shared signals only (`code/hybrid_gpt_v08`; not a reproduction of the same model)—gives the same qualitative access result (2.16 vs 3.65 shared signals, $p_{\text{true}} - 0.047$), public disagreement with private revision under cost (78.9% vs 8.7%) and zero label value for next-act prediction. On 480 dialogues its labels raise the posterior on the true profile by +0.026 [0.015, 0.036] and cut arrival log-loss by 0.073 nat, while argmax accuracy is unchanged (+0.013 [−0.023, 0.048])—metrics distinct from the recovery rate above and not pooled with it. With ω on shared signals only, its attenuated speaker is nearly indistinguishable from an honest one. (2) In v0.7 a regime variable adds nothing to next-act forecasting over the last act; hysteresis appears in *positions*, not acts. (3) The engine loses to a direct prompt, and a weak keyword baseline does not establish the gates’ value. (4) Labels are valuable only under a stipulated consequence; real-dialogue annotation must *show*

that restriction reduces verification. (5) Reciprocal amplification is weak: two access protectors widen the belief gap by only 0.017 [0.008, 0.025] over two honest agents; in v0.7 superadditive non-concession is reliable only for two attenuated agents (+0.054) and untestable for two costly ones (single agent at ceiling, 0.998).

Active inference. The contribution is an explicit generative specification with matched observation and intervention comparisons [5,4,14]; precision is a model parameter, not a neural quantity inferred from speech. The epistemic term is bounded (17% of the asking contrast in Part II; the dyadic separation survives $\beta = 0$, Appendix C); the results support comparative model development, not confirmation of active inference.

Limits. The benchmark consists of curated monologue fragments with incomplete provenance and no label-level inter-rater data [2]; the engine is not runnable from the released material. The effect of a form on evidence access is stipulated, as is the transport of two confusion rows to synthetic dyads; agents model each other without forms and without recursion; c is paid on every CONCEDE and γ prices contradiction of the initial position (the re-implementation above prices both relative to the current position and gives the same qualitative access result); $N = 2$ private records is small; nothing is fitted to people, and no maintenance or allostatic cost is modelled [16,19]. Alternative routes—prior strength, selective sampling, ordinary distrust of a source [10,26]—are not separated here.

What to measure next. The dyad tells a human study what to record: private belief before and after (separates uptake from expression; 17 dialogues per group in simulation), whether each move invites or restricts verification and how much verification occurs (separates access; 12 per group), the partner’s estimate of the speaker, and evidence exposure under a randomised incentive. The decisive test for the annotation is whether invite/restrict coding of real dyads predicts subsequent verification and belief change beyond action counts, on a corpus sampled independently of the ontology, coded by expert and lay annotators under a frozen manual with reliability reported per evidence tier, and compared against direct-prompt and rule baselines with a common output set [3,18,1]. Regime and physiological hypotheses [23,22]: Appendix D.

6 Conclusion

Protective speech is a testable proposal when the protected expectation, the observable form of the moves and the mechanism are specified separately. A typed annotation gives explicit, unevenly reproducible observables; a single-agent model shows that public non-concession under-determines the mechanism; a dyadic model connects the two and yields a third, publicly invisible mechanism with one cheap measurement for each. What remains is to measure, in real dyads, private belief, invitation and restriction, and how much gets checked.

Data and code availability. Code, archived predictions (identifiers and labels, not texts), result tables, figures and LaTeX sources: https://github.com/muqali2816/mindloom_IWAI_2026_speech_protects_model. The Part II simulator is quoted from the author’s archive, not re-run. Rater matrices, benchmark texts and the runnable engine are not released; exact reruns of the July 2026 model calls are impossible (sampling settings and provider records were not retained, Appendix B), re-scoring needs no model calls.

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A Part II: formal details

Free energy and update. Given the previous posterior Q_{t-1} and the new observation e , $F[Q] = \mathbb{E}_Q[\ln Q(x) - \ln A_{\omega,\lambda}(e \mid x) - \ln Q_{t-1}(x)] = D_{\text{KL}}[Q \parallel P(x \mid$

$e)] - \ln P(e)$; its minimiser is the exact posterior, and equal normalisation constants of the two likelihood rows yield (2).

Expected information gain. $I_\omega(q_t, u) = \sum_e P_\omega(e \mid u) D_{\text{KL}}[Q(x \mid e, u) \parallel Q(x)]$, mutual information under the subjective model at the arrival rate the action produces (λ_1 after ASK, λ_0 otherwise); an earlier build reused the previous action’s rate for some candidates, which was corrected on 28 September and all results recomputed.

Ablation limit. Removing I_ω gives the loss structure $k(1 - q)$, $kq + c$, b , b of the archived four-action simulator; the policies coincide to floating-point tolerance after renormalising over the four shared actions.

Preferences. The cost term can be written as $C(u, x) \propto \exp[-\ell(u, x)]$, giving expected negative log preference up to an action-independent constant; ℓ is not a separate KL risk term.

Recovery detail. Seed 20260929; mean posterior on the correct family for attenuation from actions alone 0.590; with the observer assuming half the true error cost, correct attribution falls to 0.07 for attenuation and rounds to 1.00 for cost; doubling the assumed temperature gives 0.13 and 0.98.

Observer. The evidence-observing observer evaluates conditional action likelihoods using the known evidence sequence; the action-only observer filters exactly over truth and cumulative signed count. Under the joint 19×16 grid prior, 0.888 of the mass already lies where both mechanisms are active; posterior mass of 0.89–0.93 in the mixed-generator checks is therefore little movement from the prior.

B Provenance, robustness and run configuration

Table 7. Benchmark construction, recomputed from the case files. Sources: Tier A `ood_regime_200`; Tier B `defense_4mon`, `ood_defense_200`. Near-duplicates: word 5-gram Jaccard ≥ 0.3 .

tier	cases	document type	words min/med/max	exact dup.	near-dup. pairs
A	169	monologue fragment	35 / 50 / 71	0	0
B	420	monologue fragment	1 / 15 / 47	0	15

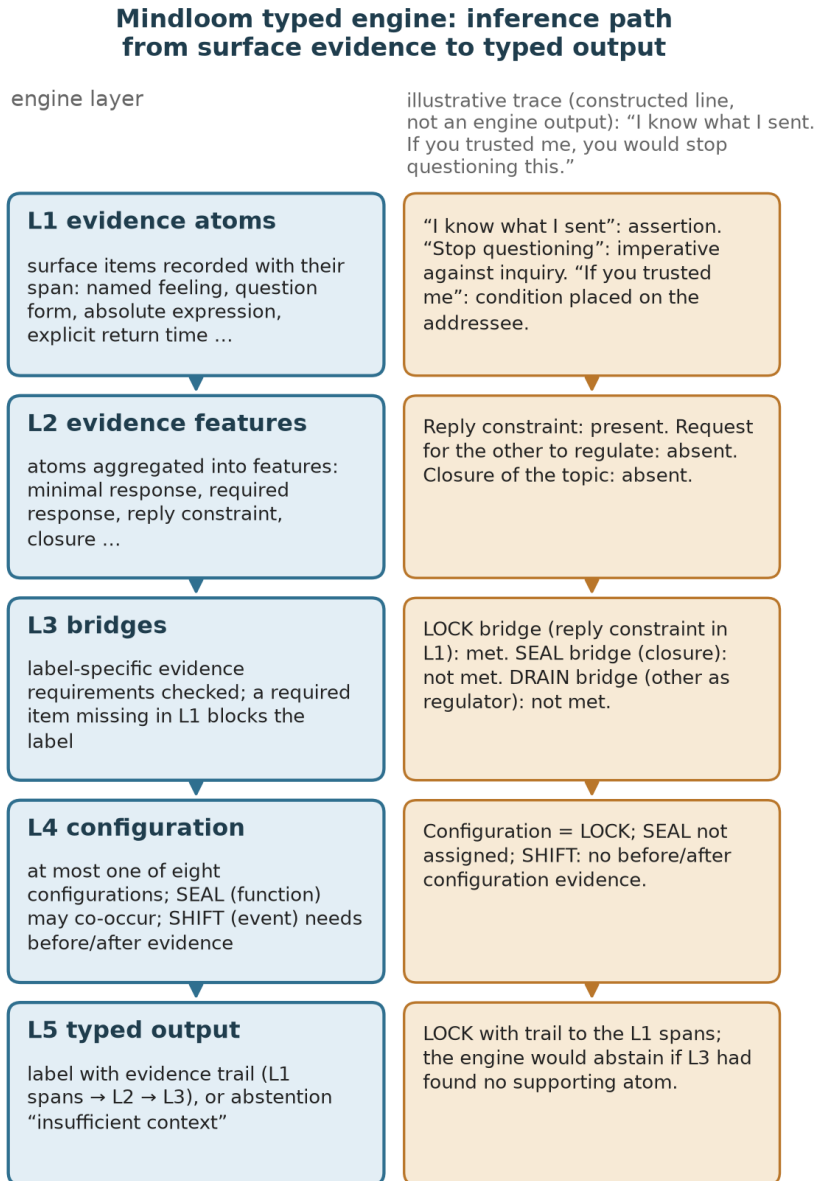


Fig. 3. Inference path of the typed engine from L1 evidence atoms to L5 typed output. Left column: the five layers; a label whose bridge finds no supporting atom in the segment's own L1 record is blocked at L3, and L5 abstains if no configuration survives. Right column: an illustrative trace for a constructed line of trajectory B (Sect. 1)—not an engine output and not a benchmark case.

B1 Annotation history. The May 17 build’s per-case medians gave 134/169 (0.8725 non-SHIFT); July label-majority scoring gives 129/169 (0.8523); build and aggregation both changed, so the historical 87% must not be presented as the July engine’s performance. A bundled May 16 rule-and-gate change reduced spurious manipulation emissions on 30 negative cases from 9 to 3 case-runs while mean strict accuracy fell from 0.744 to 0.600. A single Sonnet 4.6 run and a single Haiku run on 589 cases both reached strict primary accuracy 0.669 but differed with allowed alternatives (0.840 vs 0.899).

B2 Run configuration (as far as the archive records it). Engine: typed pipeline L1–L5, three stored runs per case, July 2026, model not recorded in the combined JSON (forensic note: Claude Haiku 4.5), temperature not recorded; abstention allowed. Direct prompt: Claude Haiku 4.5, three runs, 1 July 2026, forced choice among ten labels, prompt text in the archive; temperature not recorded. Keyword rules: deterministic; majority and random baselines: seed 0. Because temperatures and prompt hashes were not archived, exact reproduction of the model calls is not possible; only re-scoring is.

B3 Part II robustness. Across 36 settings of temperature, reliability, yield after ASK and error cost (recalibrated on 2000 worlds, tested on 2000 fresh), cost agents keep higher p_{true} and asking at every setting; the attenuation-minus-cost belief gap has median -0.24 (range -0.34 to -0.08); non-concession mismatches are below 0.02 in 33 settings. Two-step planning re-matched at $(0.10, 0)$ and $(1, 1.5)$ keeps gaps of -0.275 and -0.105 . With PRESSURE cost $b - gc$, $g = 0.5$, the re-matched cost is 1.1 and the asking contrast becomes -0.013 while the pressure contrast becomes 0.411.

B4 Reproducibility status. The author’s Part II package reports ten passing checks and a re-execution of eighteen archived v0.5 conditions (eight bitwise, ten within 7.2×10^{-13} in log-odds; 30 result tables within 10^{-10}); neither was repeated for this paper, whose Part II numbers are quoted from that package. Re-scoring is reproducible without model calls; human pilot figures are transcribed.

C Hybrid dyad: specification, validation, sensitivity

Parameters. Rounds 12, tail rounds 9–12; $\rho = 0.75$; own prior 0.75; $N = 2$; trusted precision of public factual acts 0.70; $k = 2$, $b = 1.10$, $\tau = 0.20$, $\delta = 0.12$; $\lambda_{\min} = 0.05$, $\lambda_{\text{base}} \in \{0.35, 0.90\}$; $\beta = 1.5$; grids $c \in \{0, 0.8, 1.6\}$, $\omega \in \{0.2, 0.6, 1\}$, $\gamma \in \{1, 2, 4\}$; 2000 worlds per condition, seeds in `protocol.json`. Simultaneous moves; the second-mover variants (AB/BA) are implemented and used only for the v0.7 regression.

Calibration. The protocol asked to match the three mechanisms on late non-concession to the attenuated speaker (0.981). Cost reaches it ($c = 1.2$, 0.986); access does not: even $\gamma = 8$ gives 0.833, because γ acts on forms, not on concession. Matching was therefore abandoned for access, and the protocol-typical values (0.2, 1.6, 2) are reported, with the mismatch stated.

Epistemic term. The signal term is $\lambda I(x; e)$ at tempered reliability $\rho_\omega = \rho^\omega / (\rho^\omega + (1 - \rho)^\omega)$. Against the expected entropy reduction of the implemented update (exact Bayes, then marginal shrink) over $q \in [0.02, 0.98]$: identical at $\omega = 1$; maximum absolute difference 0.019 nat at $\omega = 0.6$ and 0.014 nat at $\omega = 0.2$, where the two curves are not even positively correlated (`results/IG_heuristic_vs_shrink.csv`). With $\beta = 1.5$ this bounds the policy log-odds error at 0.14; the $\beta = 0$ run shows that no reported separation depends on the term. The term is therefore an approximation, not the information of the declared process.

Validation checks (ten, all pass): regression to the prototype’s four hypotheses and interventions; regression to the v0.7 latent dynamics at $\beta = 1.5$; $\omega = 0$ produces no belief change; large c produces no unrelieved concession; role-swap symmetry; brute-force path enumeration equals the filter (3.6×10^{-15}); null labels give zero gain; coupling off gives zero gain; scalar and vectorised policies agree; seed repeatability.

Sensitivity. 54 settings ($\tau \in \{0.1, 0.2, 0.4\}$, $\delta \in \{0.06, 0.12, 0.24\}$, $\lambda_{\min} \in \{0.02, 0.05, 0.10\}$, $N \in \{2, 4\}$) on 1 000 worlds each: no sign flip in access–costly restriction rate (range 0.07–0.41), access–honest signals (−1.31 to −0.16), access–honest p_{true} (−0.058 to −0.004), costly–attenuated belief change (0.12–0.19). Label gain at 24 corner settings: 0.033–0.084, all intervals positive. With $\beta = 0$ the three-channel separation persists; β mainly raises ASK rates.

Label channel. Gold-LOCK row: LOCK 10, SEAL 3, BUILD 2 (of 15); gold-SEEK row: SEEK 20 (of 20). Generator: one Dirichlet(counts+0.5) draw per row per world. Observer: fixed rows $(1 - \eta) \bar{C}_f + \eta \bar{C}$, with $\bar{C}_f$ the smoothed mean row and $\bar{C}$ its average over forms, $\eta \in \{0, 0.25, 0.5, 1\}$; row uncertainty is not integrated by the observer. Ratio of label to oracle signal gain: 0.59–0.67 at $\eta = 0$, 0.53–0.61 at $\eta = 0.25$, 0.38–0.47 at $\eta = 0.5$, 0 at $\eta = 1$.

Independent re-implementation (`code/hybrid_gpt_v08`). Eight rounds of sixteen alternating moves; acts STATE_0/STATE_1/ASK/SILENCE/REASSURE $\times$ form; c on a change of the current position, γ on the position after the move; $\rho_\omega = \sigma(\omega \logit \rho)$ on shared signals only; exact $I[(x, s_j, \theta_j); (E, Y_j)]$ with $\beta = 1$; observer with both profiles unknown on a four-point dictionary, stratified particle filter ($K = 8$). Its 13 validation checks pass and reproduce bitwise. Its 96-dialogue pilot found no detectable label gain in argmax profile accuracy; on 480 fresh dialogues (same code, `ext_observer.py`, contrasts with paired 95 % Monte Carlo intervals in `results/ext_observer_contrasts_480.csv`) argmax accuracy is still not detectably changed (+0.013 [−0.023, 0.048]) while the posterior on the true profile rises by +0.026 [0.015, 0.036] and arrival log-loss falls by 0.073 [0.063, 0.082] nat (oracle 0.119). This is a different observer, dictionary and metric from Table 6.

Reciprocal dyad (v0.7) findings used above. Costly $\times$ costly and attenuated $\times$ attenuated pairs reach the same mutual non-concession (0.996 vs 0.972) with belief gaps

0.22 vs 0.61; a regime variable adds ≈ 0 to next-act forecasting over the last act; hysteresis appears in positions, not acts; cost is recoverable from acts (0.90) and attenuation is not (0.62).

D Deferred roadmap: regimes and physiological cost

Regimes and forecasting. Collect repeated, sufficiently long exchanges with measured changes in evidence, incentives and relational context; define the unit (speaker sequence vs dyad sequence) in advance; require persistence, recurrent transitions and reproducibility; compare against last-act and recent-context baselines with held-out participants and calibrated probabilistic forecasts of inquiry, closure, repair and concession. The v0.7 result that a regime variable adds nothing to forecasting in the simulator sets the bar such a model must clear. A SHIFT annotation and a demonstrated change in a dynamical regime remain separate outcomes.

Acute regulation vs allostatic load. Time-aligned dialogue and physiological recording (cardiac, respiratory, electrodermal, perceived effort) across baseline, interaction and recovery, controlling speaking duration and movement, distinguishing continued exposure to a real demand from persistence after it has diminished. The programme predicts that protection may reduce immediate discomfort while prolonging regulation, and permits the opposite. Allostatic load requires a separate longitudinal design with a pre-specified multisystem index [19]; adding a latent variable named “allostatic load” without physiological validation would not test the hypothesis.

Human test of the two-plus-one mechanisms. A factual task with verifiable evidence, a scripted counterpart and the closed act menu, with private belief elicited before, at round 6 and at round 12; random assignment at round 7 to retain or remove a concession penalty; recorded evidence availability, receipt and requests; coding of each move as inviting or restricting verification. The archived protocol is an unregistered draft (illustrative power 0.95 at 80 per arm under the model’s effect, 0.52 at half); a pilot should precede registration.